4-comma: Sketch AI

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Fig. 1. Sketch AI 4-comma:AI restores a child's creativity

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The child's drawing is ruined when water is spilled on it. However, a robot offers to recreate it using AI. In this study, AI-assisted creativity is explored through DALL-E 3 and Visual ChatGPT. Artistic collaboration and co-creativity are examined. It is found that AI supports creative restoration, even for beginners.

CCS Concepts: • **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Additional Key Words and Phrases: Do, Not, Us, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 Introduction and Related Works

AI is changing how Yonkoma (4-Koma) manga is made by mixing old storytelling with new technology. Yonkoma manga started in Japan in the early 20th century and tells short but strong stories in four panels. At first, people drew it by hand, but later, it became digital. Now, AI tools help improve the process. Advanced AI models like DALL-E 3 create high-quality manga images from text[1]. This ensures all panels have the same artistic style. Also, Visual ChatGPT lets people modify AI-made images by giving feedback and making adjustments[2].AI makes manga production faster, more flexible, and more accurate. However, problems remain, such as keeping a stable visual style and making sure images fit the story. To fix this, people save AI settings and structured prompts to get consistent results[3].This study explores how AI is changing Yonkoma manga. AI automates repetitive tasks, allowing people to focus more on storytelling. This helps them be more creative while keeping the manga's unique style. As AI improves, it will enhance efficiency and artistic expression. AI will play a bigger role in digital manga and change how it is created in the future. Using AI instead of old methods will redefine manga creation.

2 Methodology

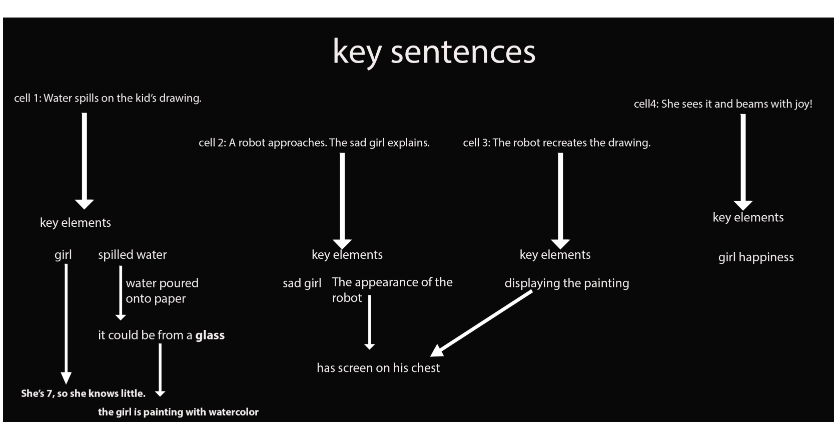

Fig. 2. key sentence for each representing a panel

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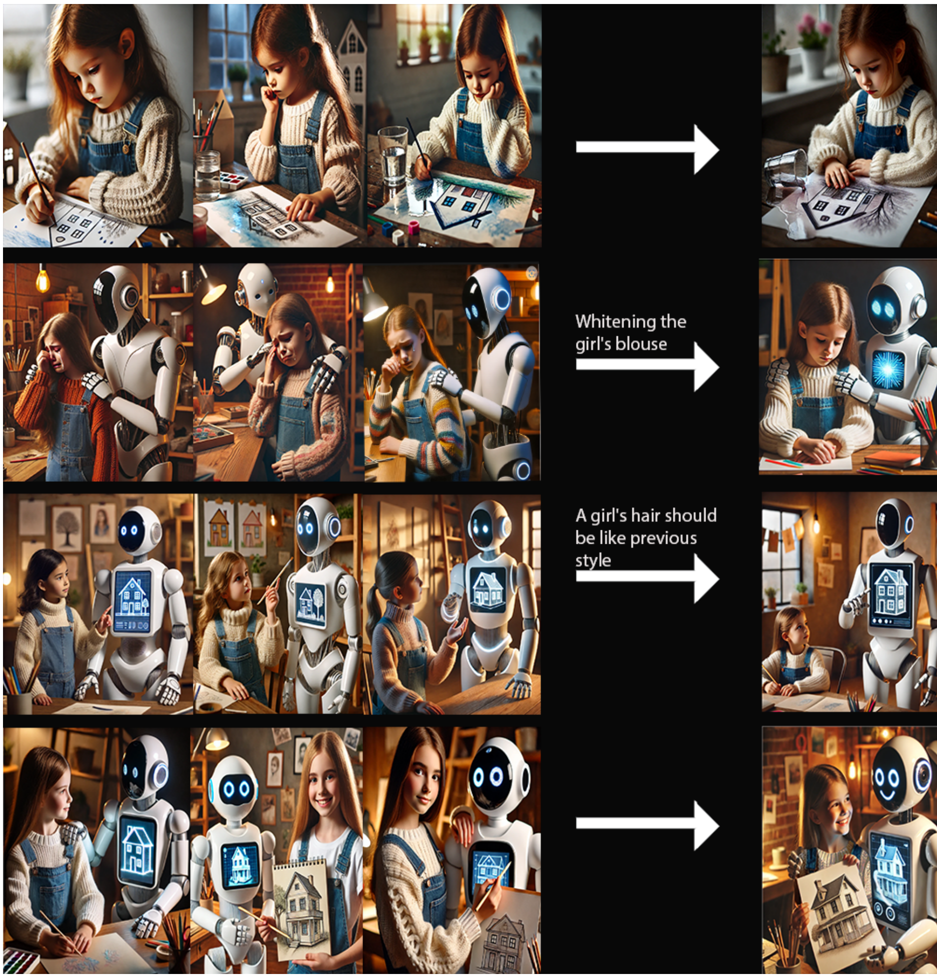

Fig. 3. From structured descriptions to refined images—DALL-E 3 generates, Visual ChatGPT perfects.

The story is divided into four sections to integrate AI-driven creativity into a 4-Koma (Yonkoma) manga, with a key sentence for each representing a panel. This structured approach helps maintain narrative clarity and allows for effective visual production. The process begins by defining and breaking the main idea into four structured descriptions. Each description includes essential details about composition, character expressions, and the overall scene setting. These structured descriptions are then fed into DALL-E 3, a model selected for its high accuracy in following text instructions and generating high-quality, visually appealing images. Additionally, Visual ChatGPT is employed to refine the generated images gradually, ensuring consistency in artistic style across all panels. This method is chosen over manual drawing because AI-based tools provide faster iterations, enhanced creative flexibility, and improved precision. AI models like DALL-E 3 can generate well-aligned, accurate images when guided by detailed prompts and structured input. To ensure that others can understand the results of this process, all AI-generated images from OpenAI were carefully documented to make it easy to reproduce the same method. This study demonstrates that blending structured storytelling with AI-generated visuals significantly enhances creative collaboration between humans and machines, paving the way for future innovations.

3 Result and Future Work

A Yonkoma (4-Koma) manga was created, successfully conveying a complete story in just a few frames. Key sentences were defined to emphasize plot points, and through interactive sessions, images were generated to maintain the narrative flow, resembling a dynamic movie sequence. If the project were to be repeated, additional AI models would be incorporated to achieve more diverse visual outputs. Detailed images were effectively produced when clear, step-by-step guidance was provided. For future projects, a more extensive story will be developed, and a short film adaptation will be created using multiple AI tools to enhance creativity and ensure artistic consistency.

4 Conclusion

This study shows how AI changes Yonkoma manga creation. The most interesting part was how AI keeps the art style the same while improving storytelling. By using clear prompts and refining images step by step, AI helps creativity. In the future, AI will expand artistic possibilities and reshape human-machine collaboration.

Acknowledgments

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